MT1011 Engineering Materials

Phase Diagrams

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Solid solutions

- All practical materials have some contamination
- Even high purity semiconductor products have some measurable level of impurity atoms.
- Commercial alloys are mixtures of different metallic elements.
- Most materials are **solid solutions**.

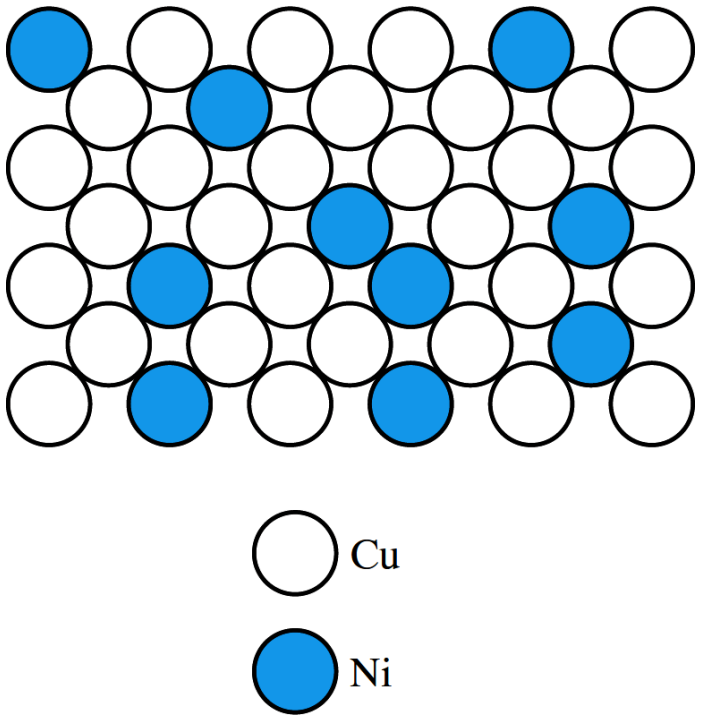


Figure: Cu-Ni solid solution.
Picture credit: Introduction to materials science for engineers/James F. Shackelford (2014)

A solid solution is similar to a liquid solutions such as the water–alcohol system.

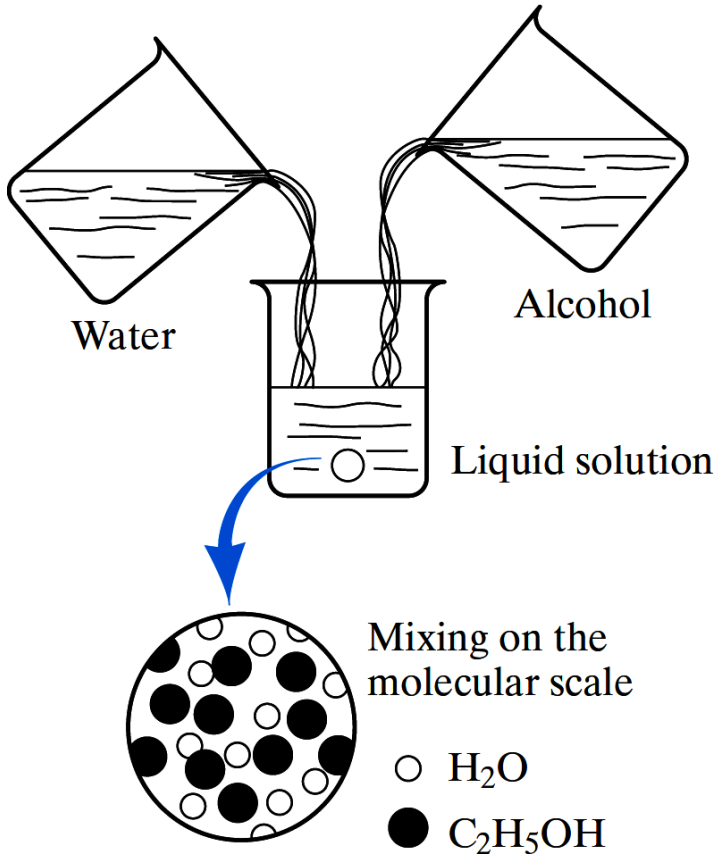


Figure: A liquid solution of water and alcohol. Mixing occurs on the molecular scale.
Picture credit: Introduction to materials science for engineers/James F. Shackelford (2014)

Solid solutions

- Solid solution of Cu and Ni atoms has the fcc structure.
- Ni acts as the solute.
- Copper is the solvent.
- This configuration is called a **substitutional** solid solution
- Ni atoms are substituting for Cu atoms on the fcc atom sites.
- This configuration will tend to occur when the atoms do not differ greatly in size.

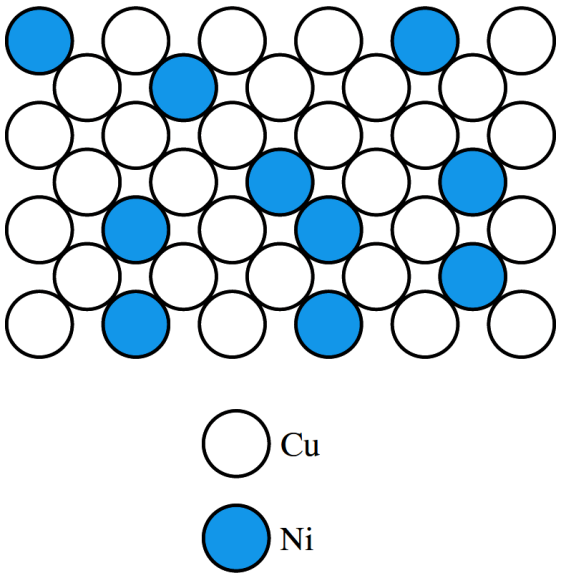
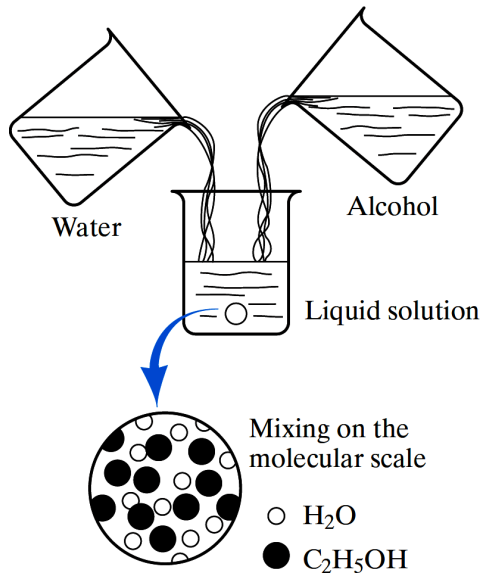


Figure: Cu-Ni solid solution.
Picture credit: Introduction to materials science for engineers/James F. Shackelford (2014)

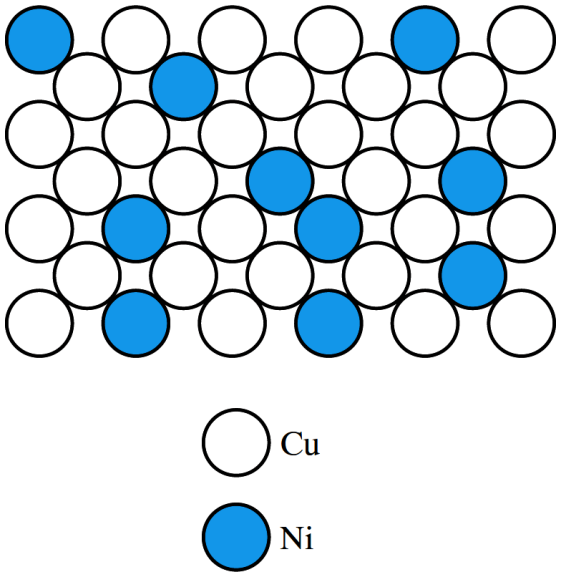
Solid solutions

- The water–alcohol system is an example of two liquids **completely soluble** in each other in all proportions.
- Cu-Ni system also completely soluble in each other in all proportions.



For complete miscibility in metallic solid solutions, the two metals must be quite similar, as defined by the Hume- Rothery rules:

- Less than about 15% difference in atomic radii
- The same crystal type
- Similar electronegativities
- The same valence



If one or more of the Hume- Rothery rules are violated, only partial solubility is possible.

At. No	Atom	Structure	At. radius (nm)	Ion	Ionic radius (nm)
13	Al	FCC	0.143	Al^{3+}	0.057
14	Si	Diamond	0.117	Si^{4-}	0.198
28	Ni	FCC	0.125	Ni^{2+}	0.078
29	Cu	FCC	0.128	Cu^{2+}	0.072
				Cu^{+}	0.096

- Only less than 2 at.% Si is soluble in Al.
- Copper is soluble in Ni at all proportions.



Phase diagrams

Phase diagrams try to answer the following question:

What microstructure should exist at a given temperature for a given material composition?

- Strong relationship between microstructure and mechanical properties.
- Microstructure of an alloy is related to the phase diagram.
- Phase diagrams provide information about processing and degradation.
- Phase diagrams help investigating failure of materials.



Component

Components are pure metals and/or compounds of which an alloy is made.

- For example, in **brass**, the components are Cu and Zn.
- For material systems with compounds, the compounds can be components.
- For example, MgO and NiO form solid solutions in a similar way to Cu and Zn in brass.



System

The series of possible alloys consisting of the same components.

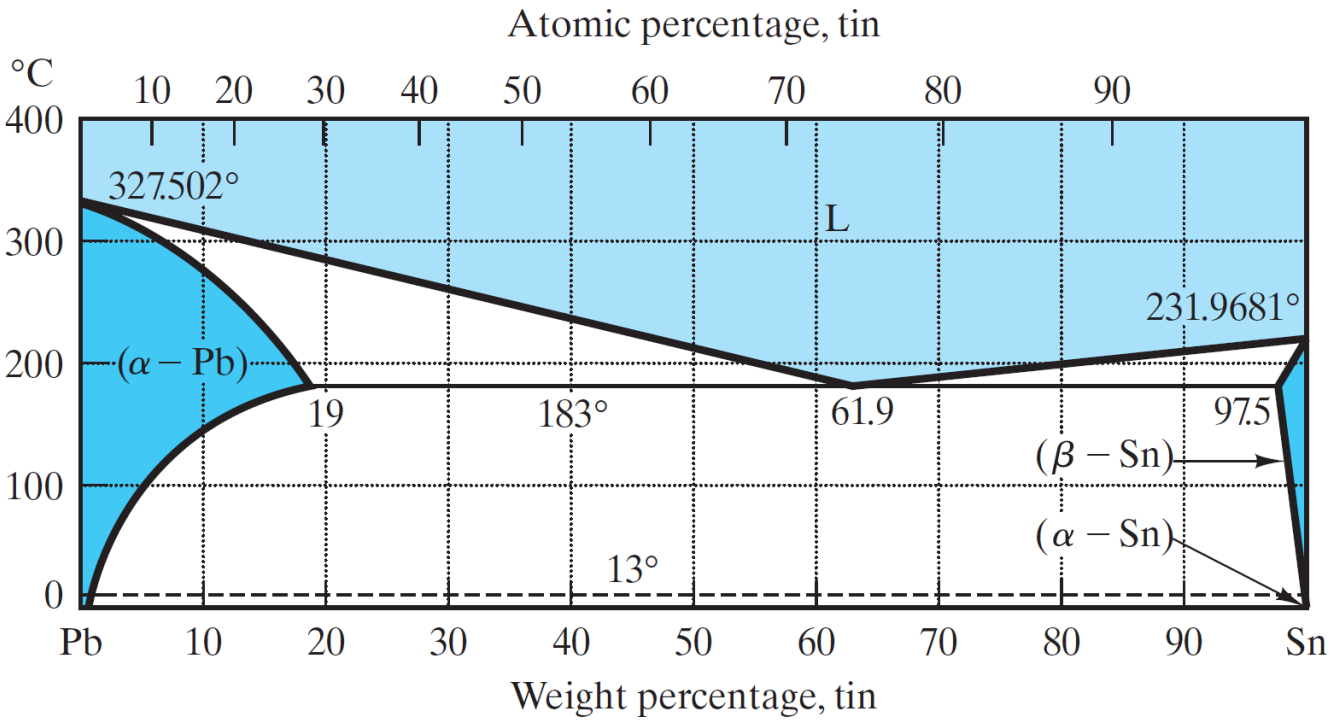


Figure: Pb-Sn phase diagram for the Pb-Sn system.

Solubility limit

The maximum concentration of solute atoms that may dissolve in the solvent to form a solid solution.

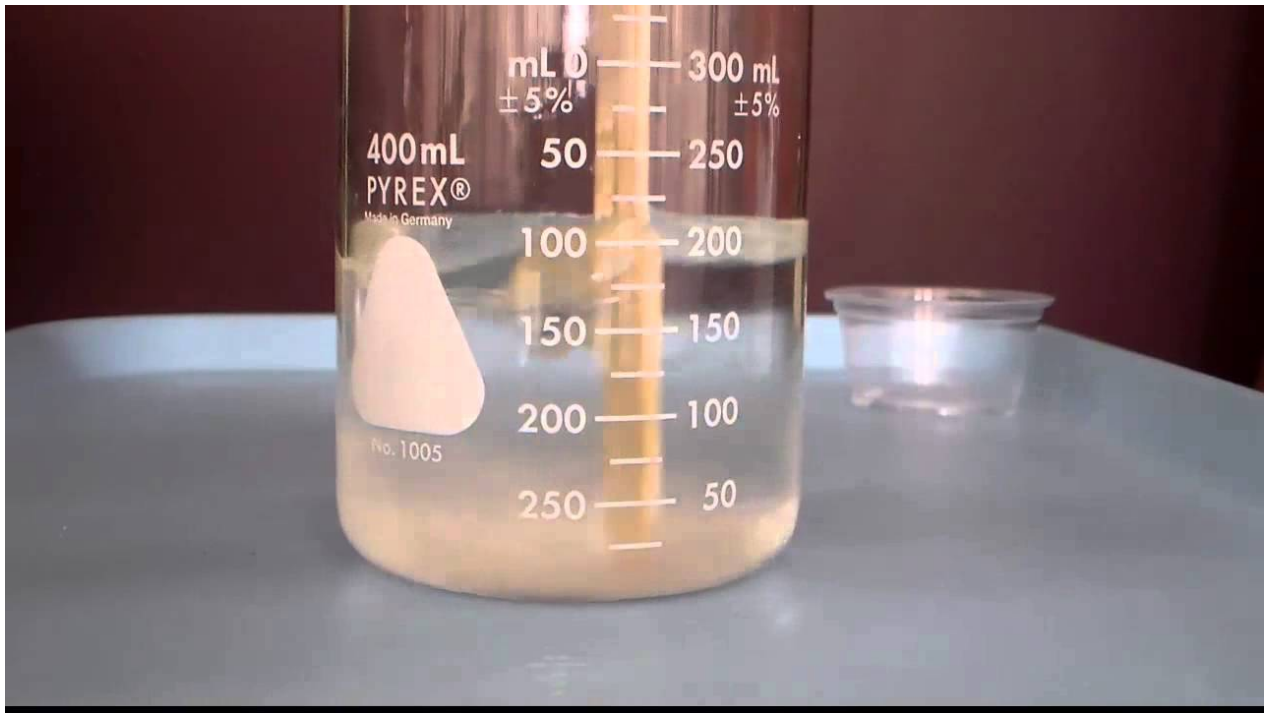
- The addition of more solute above the solubility limit will give anothr another solid solution or compound.
- In the sugar–water ($C_{12}H_{22}O_{11}$–H_2O) system, as sugar is added to water, a sugar–water solution or syrup forms.



Phase diagrams

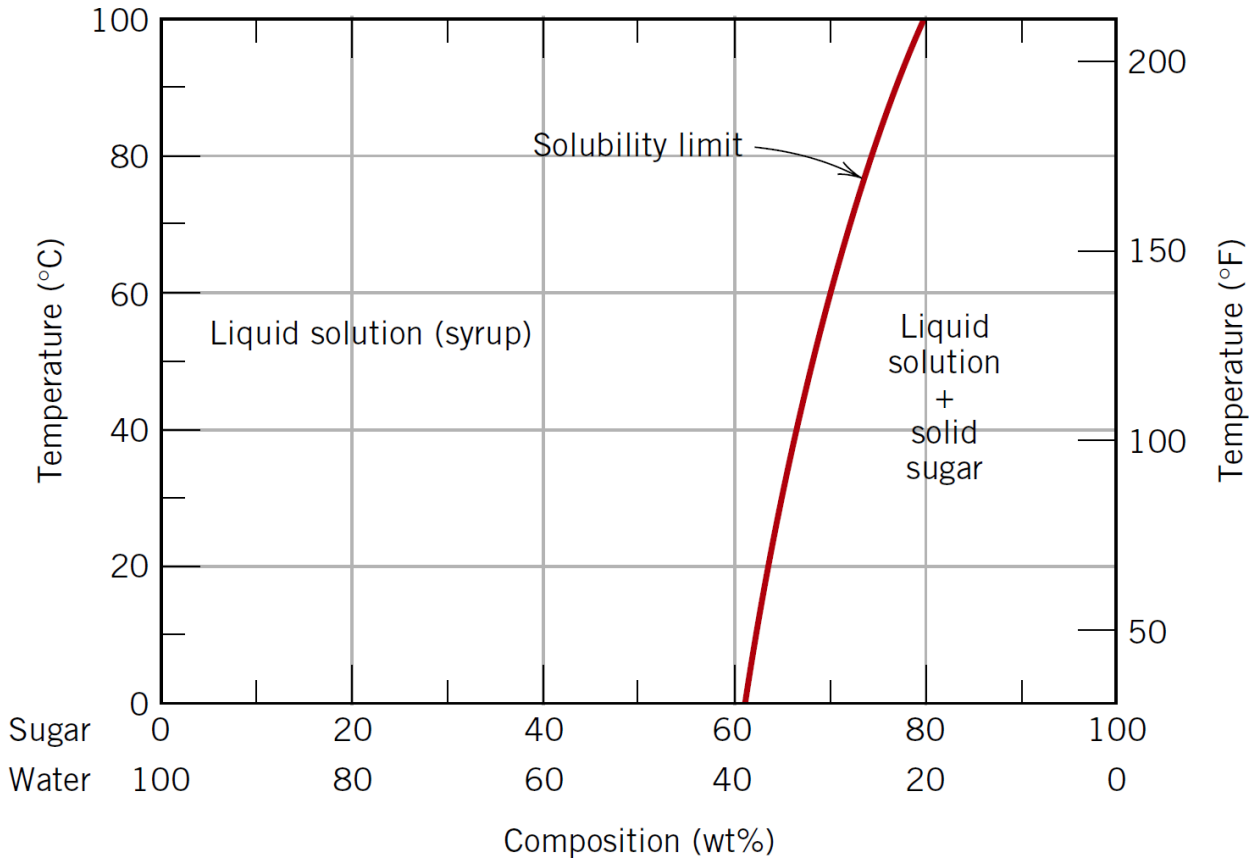
Solubility limit

- As more sugar is added, the solution becomes more concentrated, until the solubility limit is reached.
- At this point, the solution becomes saturated with sugar.
- The solution is not capable of dissolving any more sugar.
- Further additions will settle to the bottom of the container.



The system now have two separate substances:

1. A sugar–water syrup liquid solution
2. A solid crystals of undissolved sugar



Phase

A homogeneous portion of a system that has uniform physical and chemical characteristics.

- Every pure material is considered a phase.
- Every solid, liquid, and gaseous solution.
- For example, the sugar–water syrup solution is one phase, and solid sugar is another phase.
  - Each phase has different physical properties.
  - Each phase is different chemically different.
  - One phase is virtually pure sugar, the other is a solution of H_2O and $C_{12}H_{22}O_{11}$.



Homogeneous system

A single-phase system is termed homogeneous

Heterogenous sustem

Systems with two or more phase mixtures

- Most metallic alloys, ceramic, polymeric, and composite systems are heterogeneous.
- The property combination of the multiphase system is different from the individual phases.



- Physical and mechanical properties depend on the microstructure.
- Microstructure can be observed using microscopes (optical or electron microscopes).

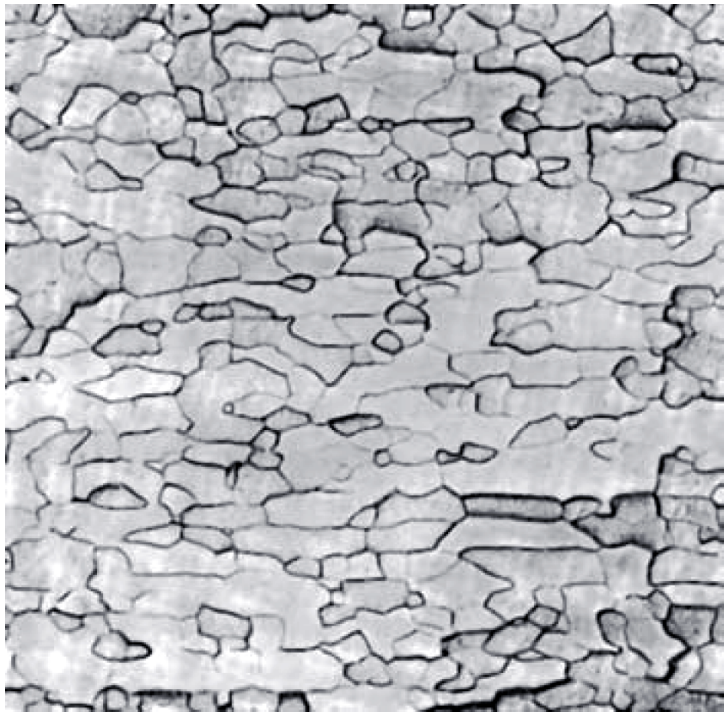


Figure: Single-phase microstructure of commercially pure molybdenum showing many grains, each grain has the same uniform composition.

Phase diagrams

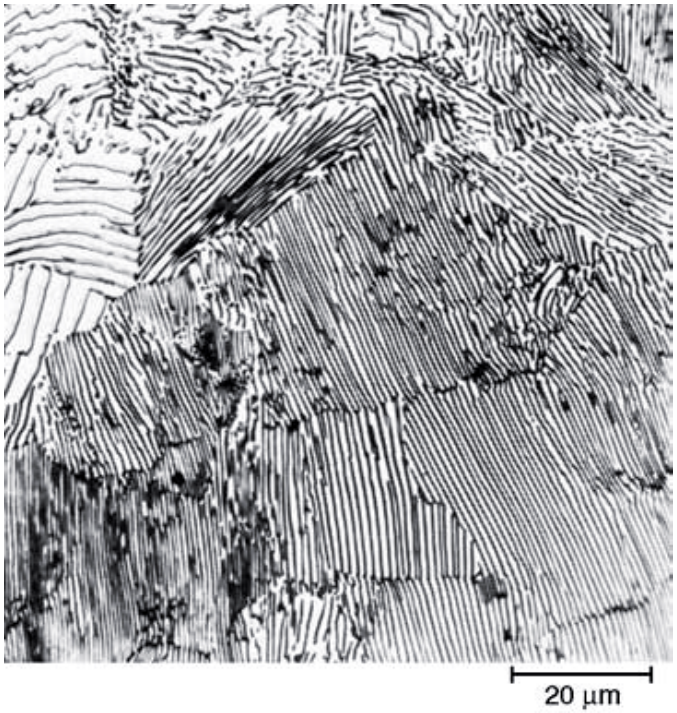
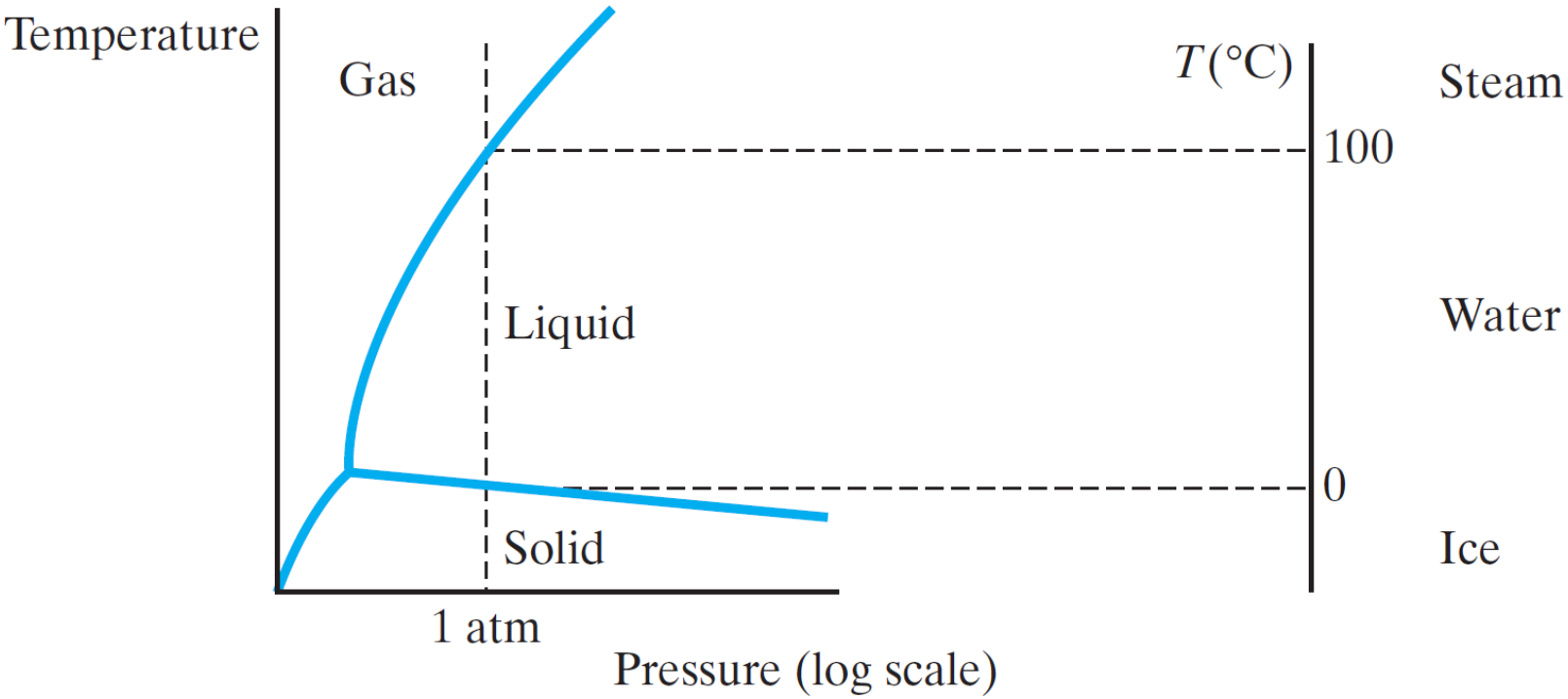


Figure: Two-phase microstructure of pearlite found in a steel with 0.8 wt% C. This carbon content is an average of the carbon content in each of the alternatinglayers of ferrite (with < 0.02 wt% C) and cementite (a compound, Fe_3C, which contains 6.7 wt% C). The narrower layers are the cementite phase.

Phase diagrams

Single component (H_2O)

- Solid H_2O (ice) transforms to liquid water at $0\,^{\circ}C$
- Water transforms to gaseous steam at $100\,^{\circ}C$.



Phase diagrams

Single component (Fe)

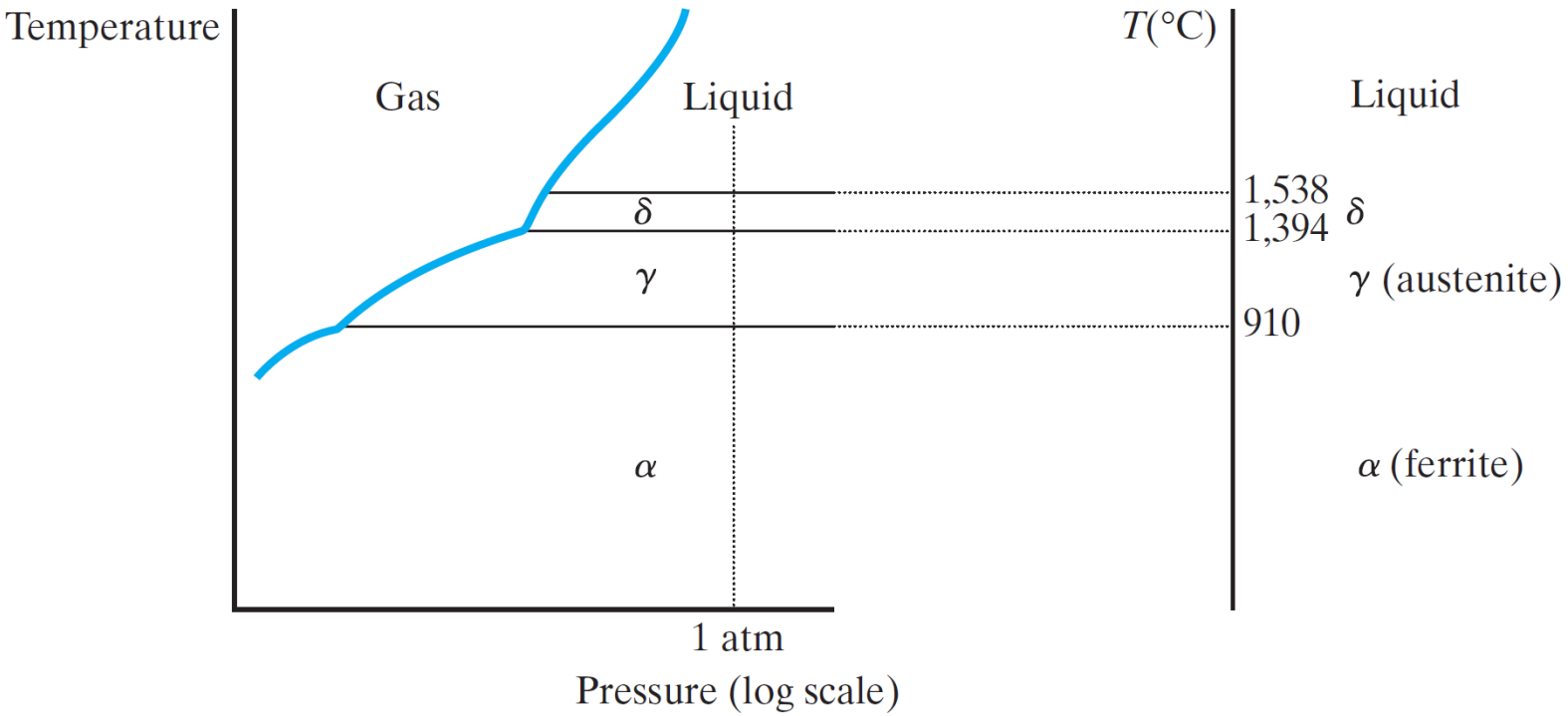
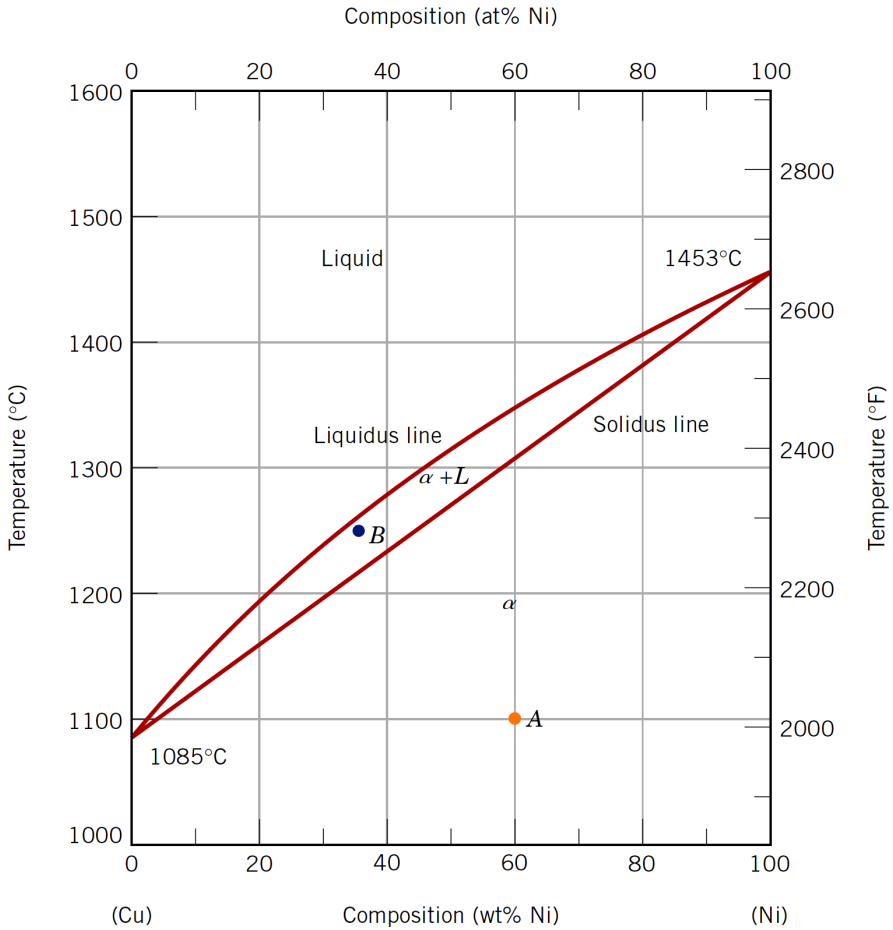


Figure: Phase diagram for pure iron.

Binary phase diagrams

Cu-Ni isomorphous system

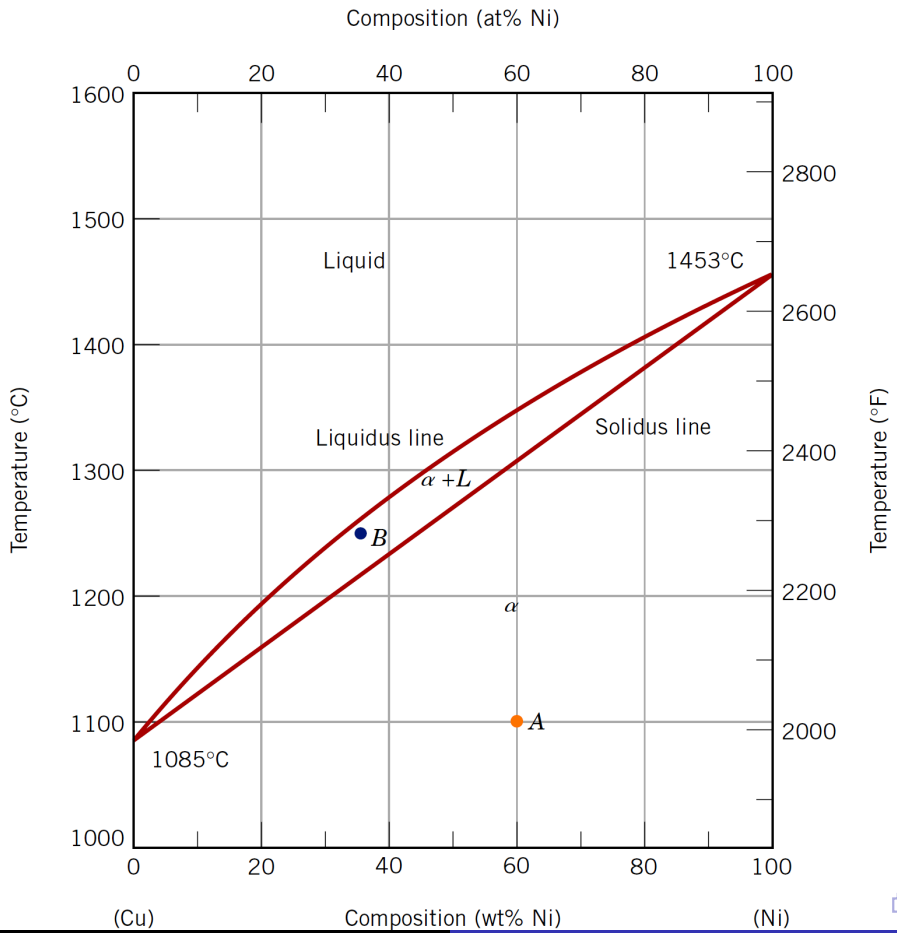
- The liquid L is a homogeneous liquid solution of Cu and Ni
- The α phase is a substitutional solid solution consisting of both Cu and Ni atoms with an FCC crystal structure.



Binary phase diagrams

Cu-Ni isomorphous system

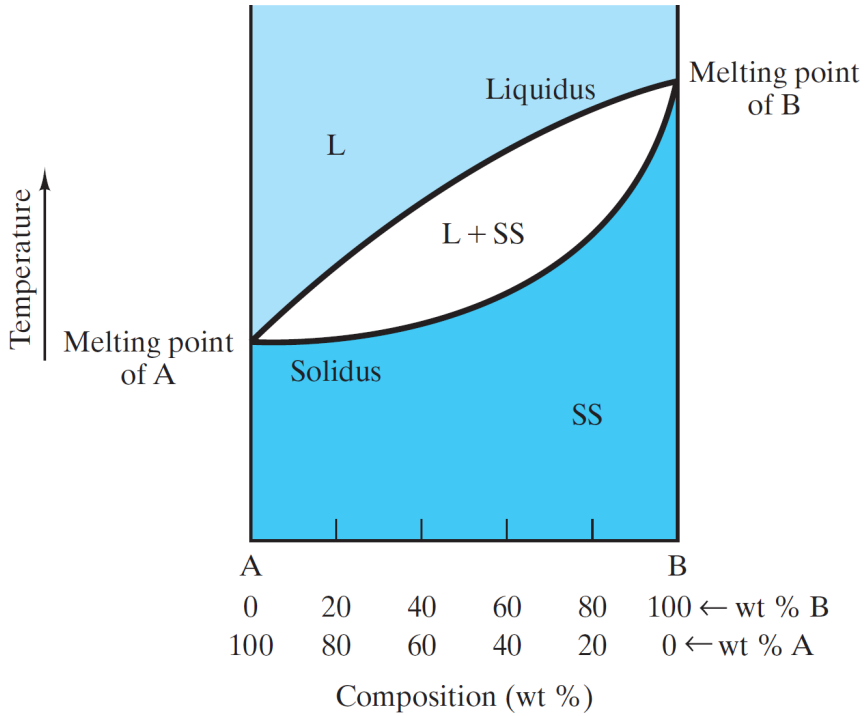
- At temperatures below about $1080\,^{\circ}\mathrm{C}$, Cu and Ni are soluble in each other in the solid state for all compositions.
- The copper–nickel system is termed **isomorphous** because of this complete liquid solid solubility of the two components.



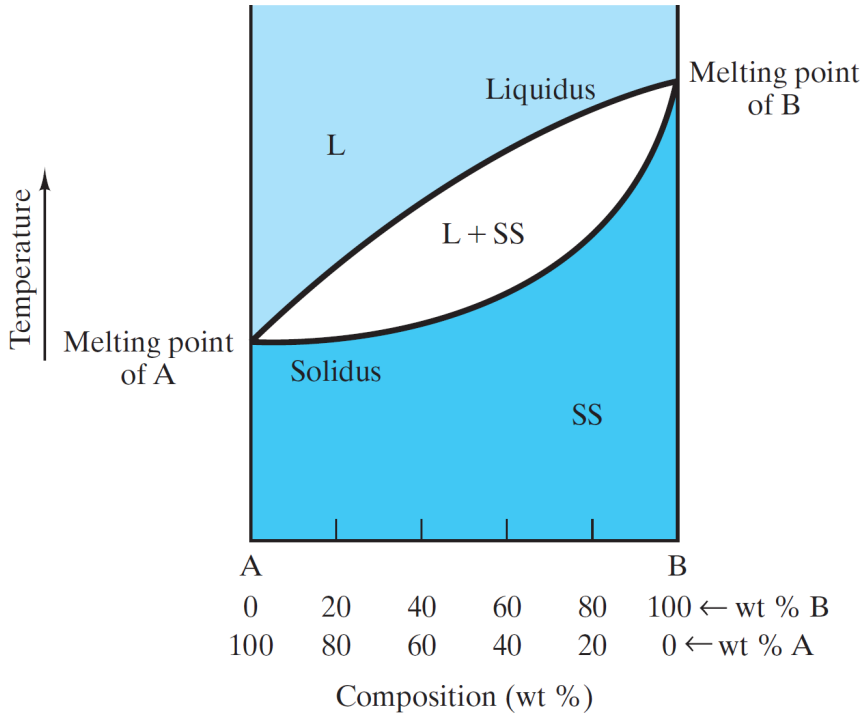
Binary phase diagrams

Isomorphous systems

- A and B are completely soluble in each other in both the liquid and solid states.
- Some systems do not have complete solubility in the liquid state, for example in the oil-water system



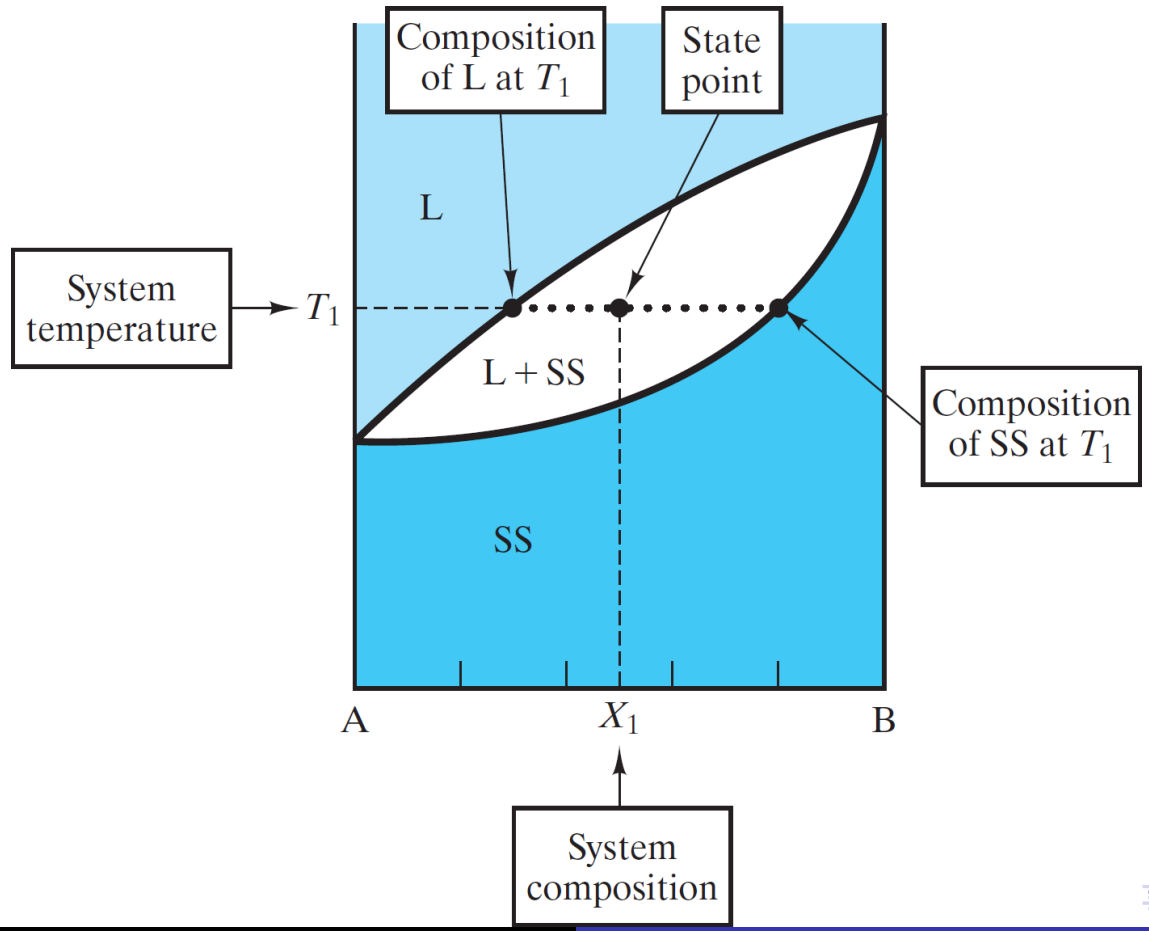
- **SS**: single solid-solution phase field.
- **L+SS**: Two phase region between the two single-phase fields.
- **Liquidus**: Upper boundary of the two-phase region. Above the liquidus the solution is liquid.
- **Solidus**: The lower boundary is called the solidus. Below the solidus the system has completely solidified.



Binary phase diagrams

Isomorphous systems

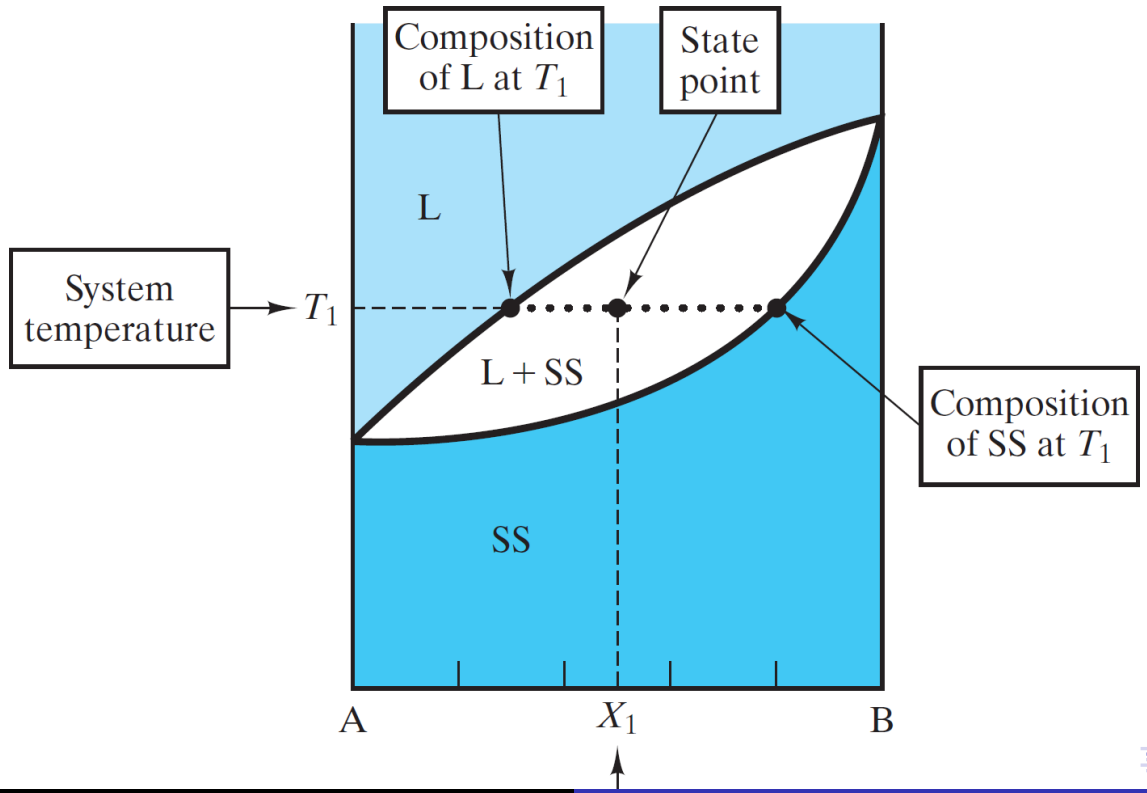
- At a given state point within the two- phase region, an A-rich liquid exists in equilibrium with a B- rich solid solution.
- The composition of each phase can be found using the horizontal passing through the state point.



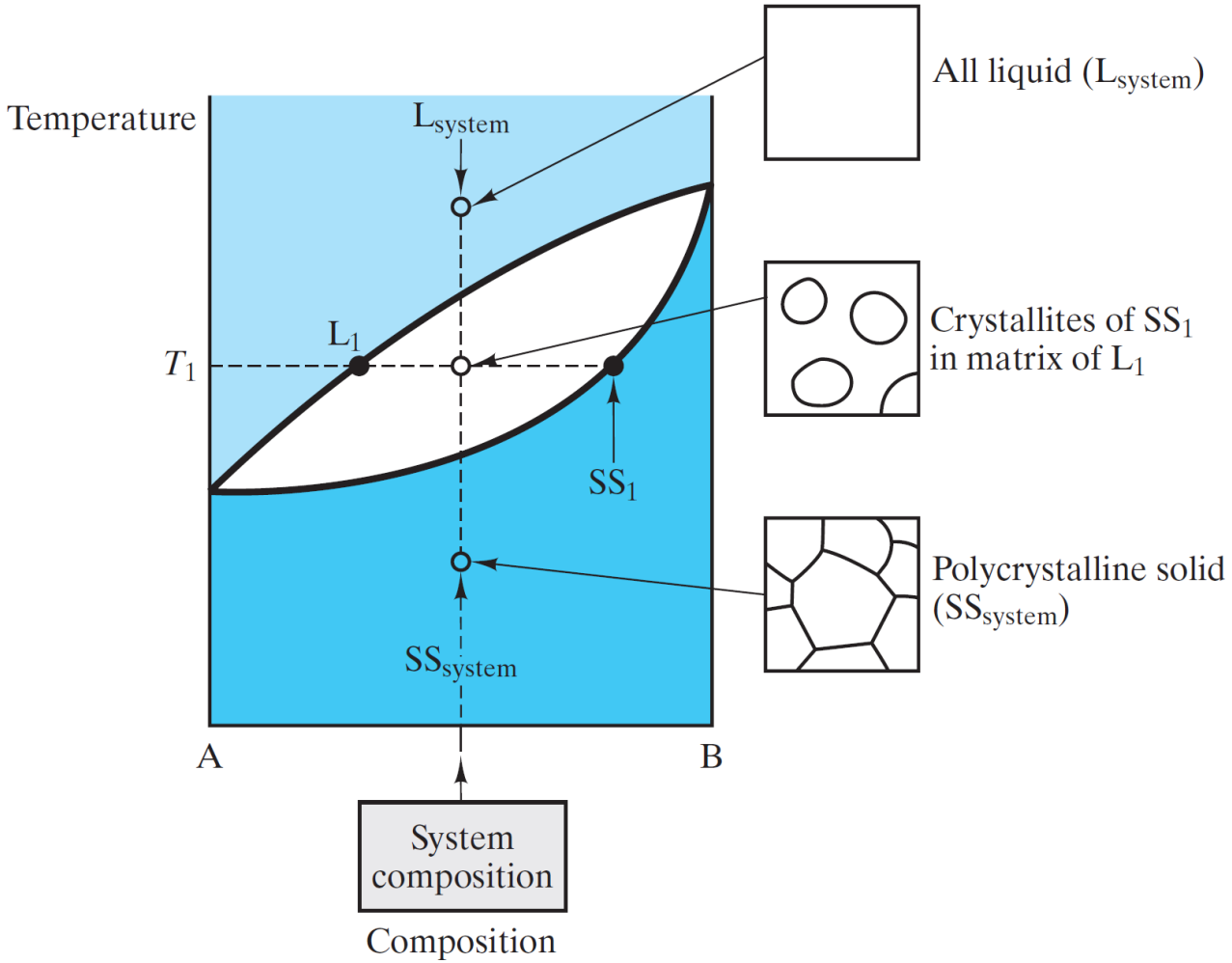
Binary phase diagrams

Isomorphous systems

- Liquid phase composition is given by the intersection point with the **liquidus**.
- Solid solution composition is given by the intersection point with the **solidus**.
- This horizontal line connecting the two phase compositions is termed a **tie line**.



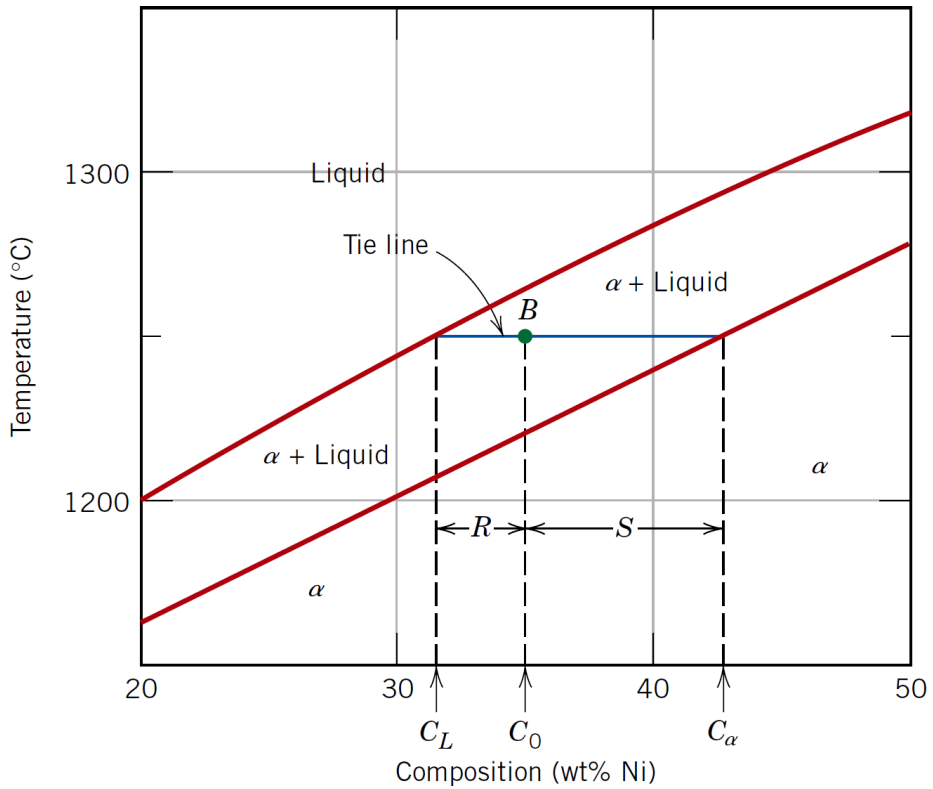
Summary of phase structure development during cooling:



Binary phase diagrams

Determination of phase composition

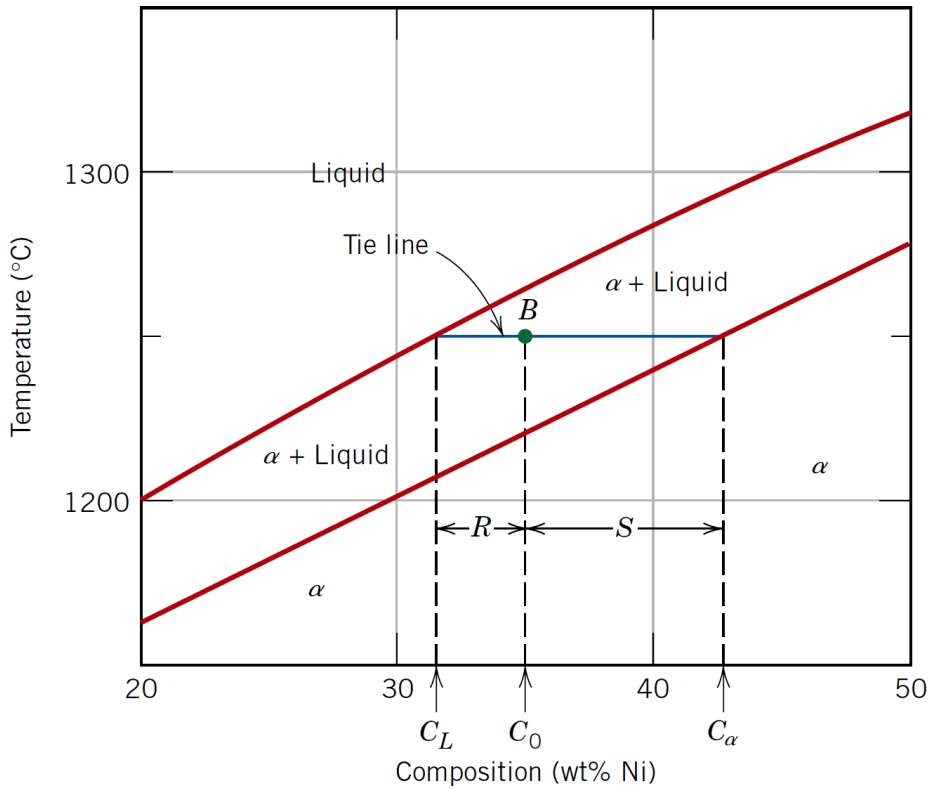
1. A tie line is drawn across the two-phase region at the temperature of the alloy.
2. Crossing of the tie line and phase boundaries are noted.
3. Perpendiculars are dropped from the intersections to the composition axis to read the composition of each phase.



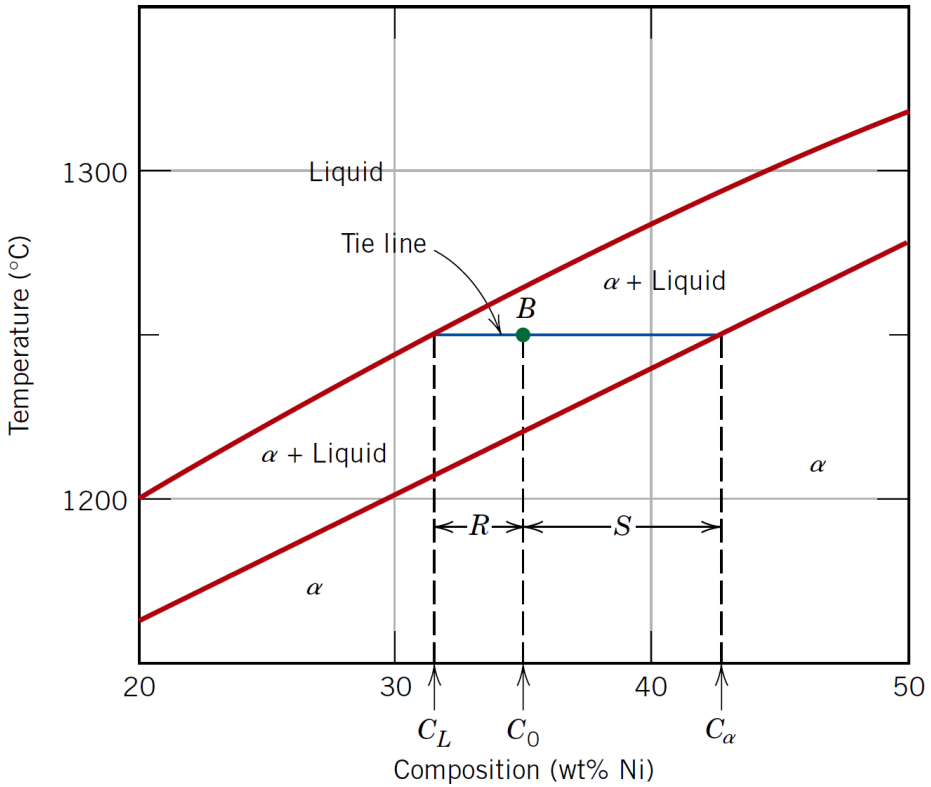
Binary phase diagrams

Determination of phase amounts

- The tie line is constructed across the two-phase region.
- The overall alloy composition is located on the tie line.
- Fraction of one phase is calculated by taking the length of tie line from the overall alloy composition to the phase boundary for the other phase, and dividing by the total tie line length.



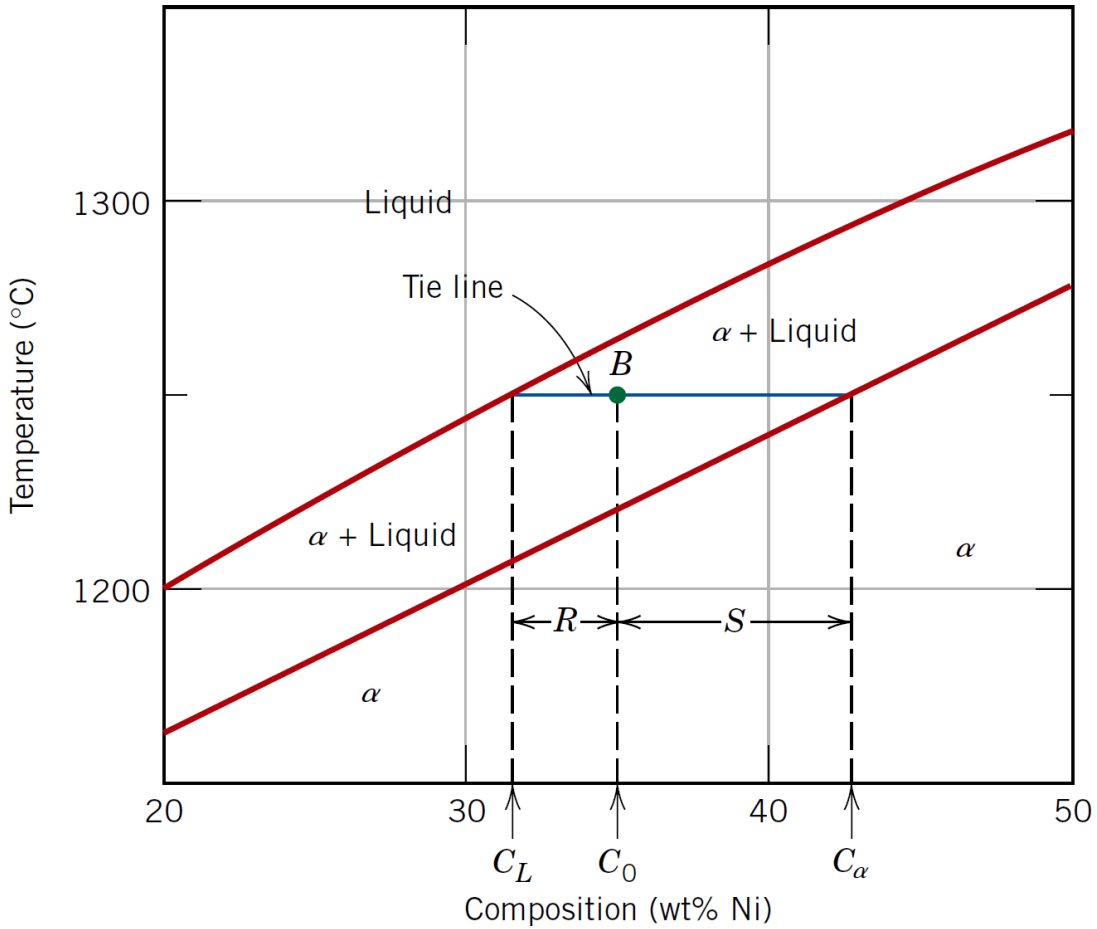
The fraction of the other phase is determined in the same manner.



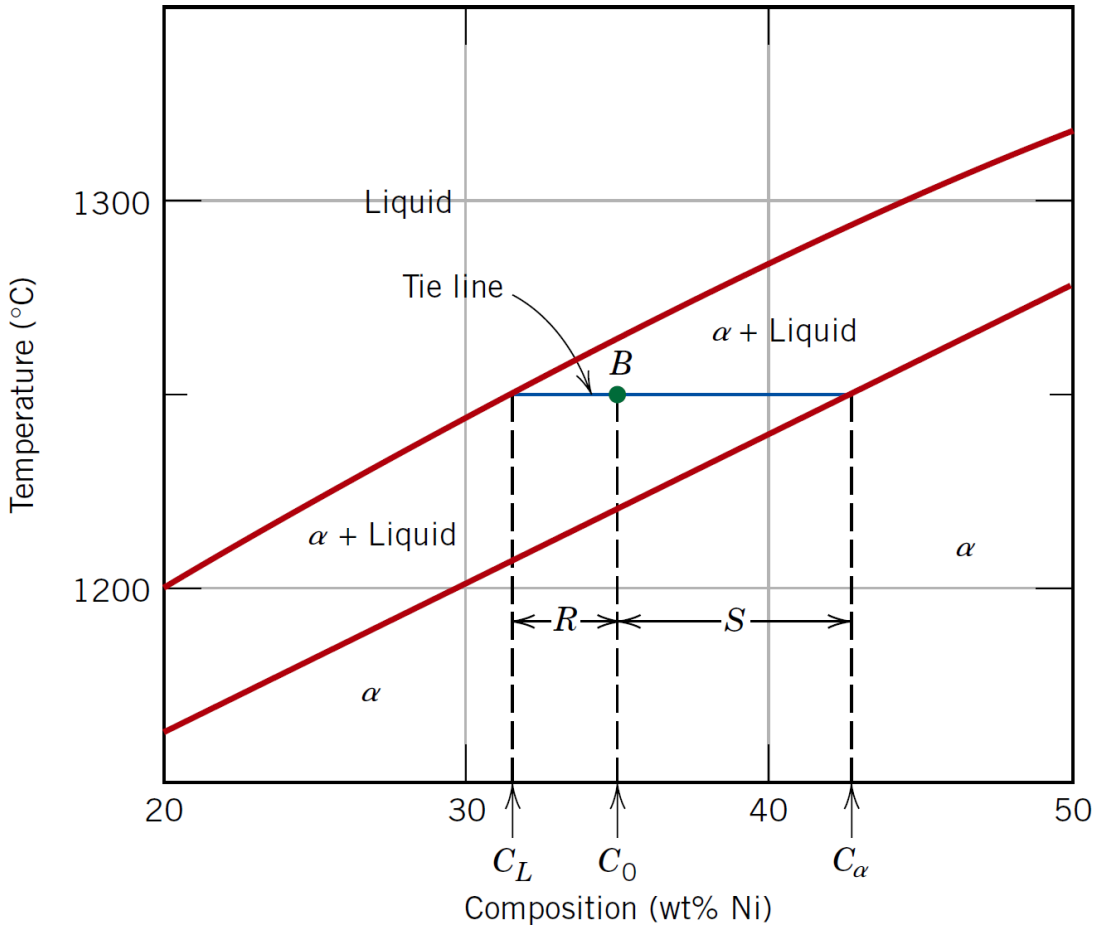
Binary phase diagrams

Determination of phase amounts

$$W_L = \frac{S}{R+S}$$

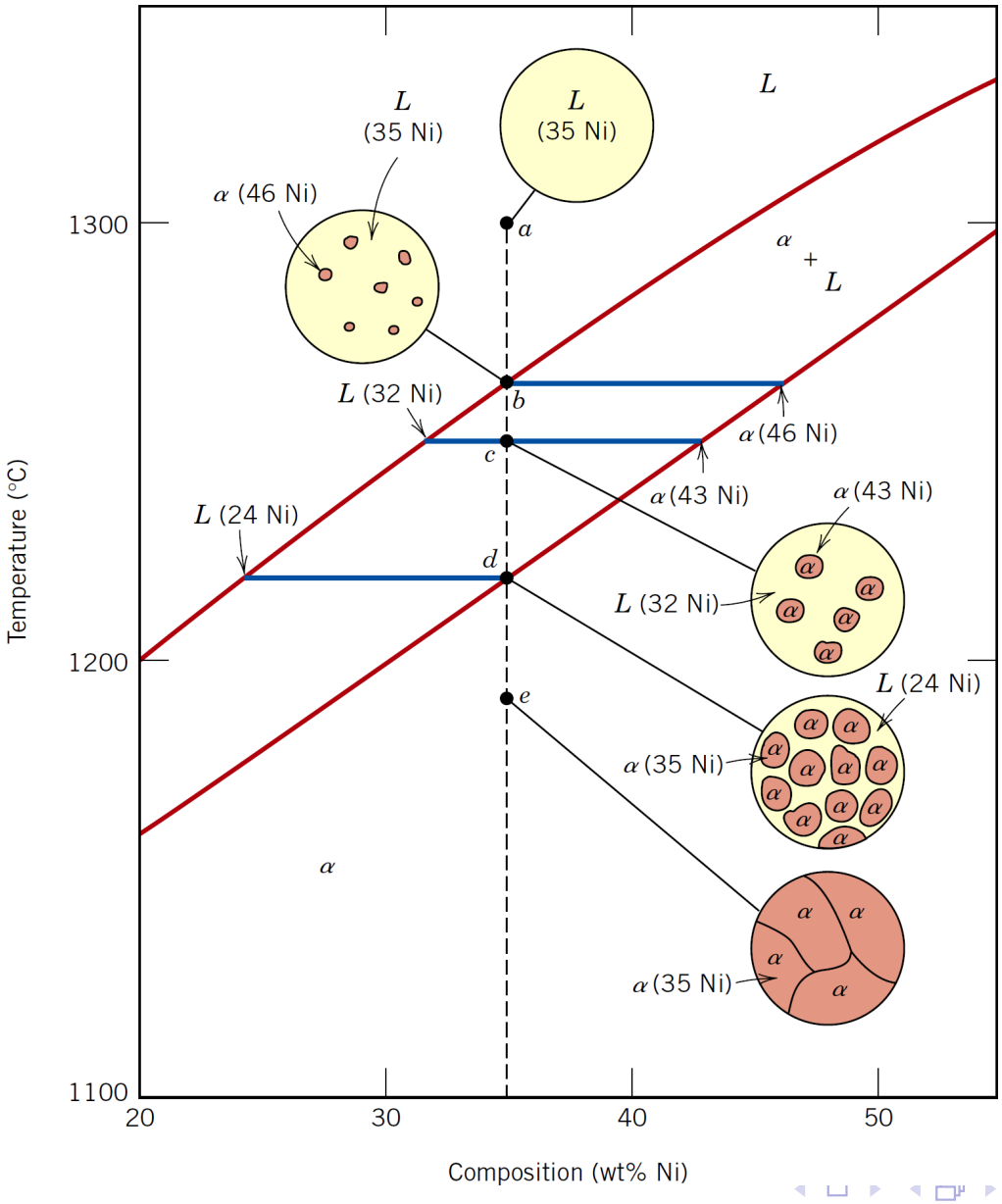


$$W_L = \frac{C_\alpha - C_0}{C_\alpha - C_L}$$



Binary phase diagrams

Equilibrium cooling



Exercise

Explain the microstructure development during the cooling of a binary alloy containing 35 wt.% Ni

Exercise

Compare the microstructure developed during the cooling of a binary alloy containing 35 wt.% Ni under equilibrium and non-equilibrium cooling conditions.

Binary phase diagrams

Mechanical properties of isomorphous alloys

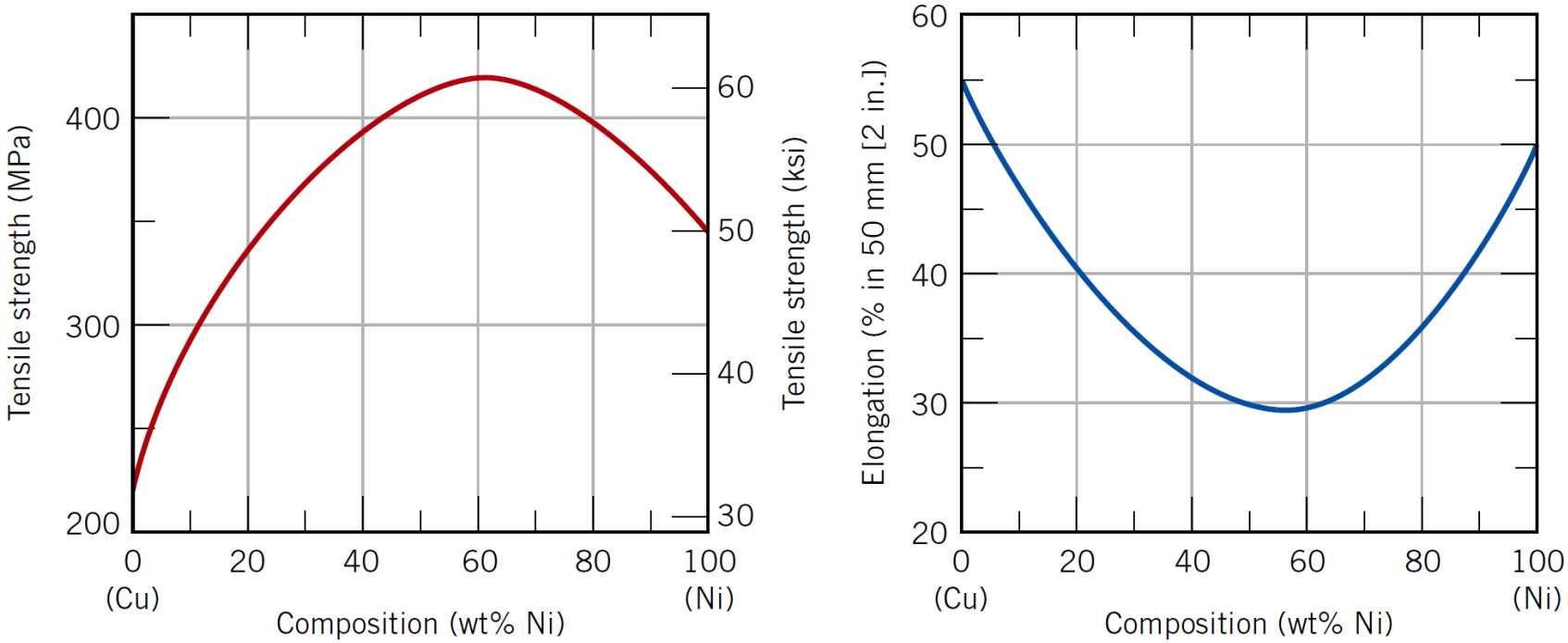


Figure: Mechanical property variation of the Ni-Cu system at room temperature.

- For all temperatures and compositions below the melting temperature of the lowest-melting component, only a single solid phase will exist.

Binary phase diagrams

Mechanical properties of isomorphous alloys

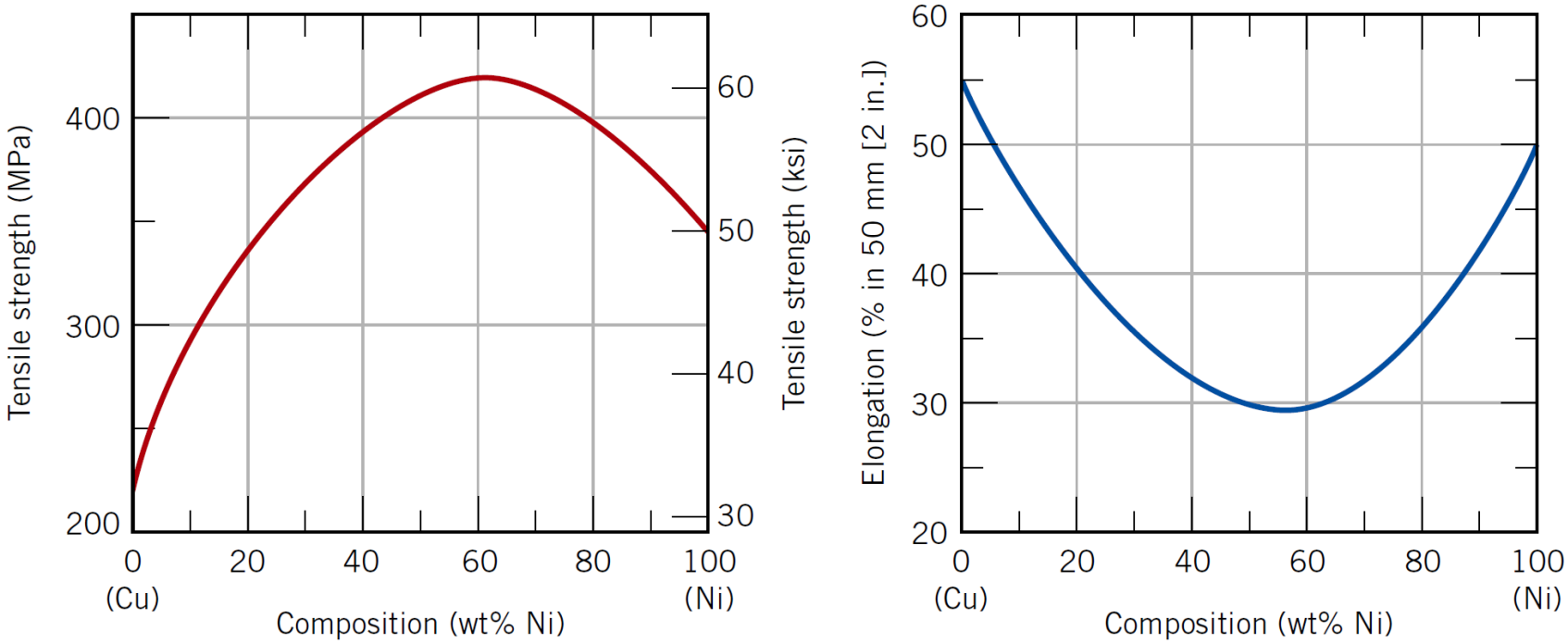


Figure: Mechanical property variation of the Ni-Cu system at room temperature.

- Therefore, each component will experience solid-solution strengthening, or an increase in strength and hardness by additions of the other component.

Binary phase diagrams

Mechanical properties of isomorphous alloys

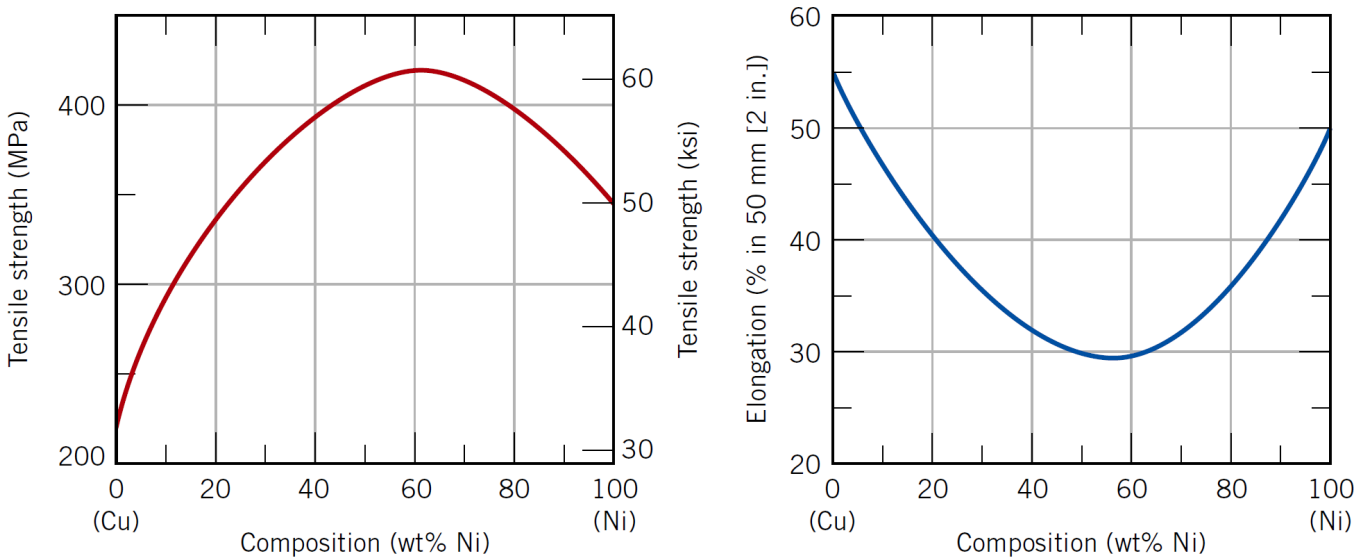


Figure: Mechanical property variation of the Ni-Cu system at room temperature.

- At some intermediate composition, the curve necessarily passes through a maximum.
- Ductility shows similar but the opposite trend (shows a minimum).